

SHIMONOSEKI, Japan

WMO: 477620

Lat: 33.9486N

Lon: 130.9264E

Elev: 20

StdP: 101.09

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.6	2.7	-8.3	1.9	3.9	-7.1	2.1	4.4	11.9	4.5	10.9	4.7	5.7	300	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	32.4	25.8	31.5	25.6	30.6	25.3	26.7	30.4	26.2	29.9	25.9	29.5	3.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.0	29.0	25.2	20.4	28.7	24.7	19.8	28.5	84.0	30.6	82.0	30.0	80.2	29.6	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.7	8.5	7.4	-0.8	33.9	1.6	0.9	-2.0	34.6	-2.9	35.2	-3.7	35.7	-4.9	36.4		
			-2.8	27.4	1.5	0.7	-3.9	27.9	-4.8	28.2	-5.6	28.6	-6.7	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	7.1	7.6	10.4	14.9	19.4	22.8	26.9	28.4	25.0	20.0	14.6	9.6
	DBStd	7.65	2.39	2.63	2.71	2.62	2.01	1.92	2.15	1.68	2.30	2.39	2.82	2.75
	HDD10.0	239	93	74	27	1	0	0	0	0	0	1	42	
	HDD18.3	1407	348	300	246	105	12	0	0	0	11	115	271	
	CDD10.0	2901	4	8	40	149	292	384	525	571	449	311	139	28
	CDD18.3	1028	0	0	0	3	45	134	267	313	199	63	3	0
	CDH23.3	8205	0	0	1	7	78	476	2551	3553	1413	127	1	0
	CDH26.7	2469	0	0	0	1	4	62	792	1321	284	4	0	0
Wind	WSAvg	3.1	3.9	3.5	3.5	3.2	3.0	2.6	2.7	2.7	2.6	2.7	3.1	3.8
Precipitation	PrecAvg	1777	79	86	115	136	157	274	317	186	175	83	93	76
	PrecMax	2287	211	146	263	251	338	483	781	432	419	225	182	135
	PrecMin	1098	26	33	55	37	36	55	63	29	63	12	30	23
	PrecStd	307	39	28	47	57	72	124	193	107	87	56	43	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	16.7	19.7	24.2	27.0	30.1	33.2	33.8	31.5	27.2	22.2	17.3
		MCWB	11.2	13.0	13.5	17.3	18.2	24.2	25.7	26.0	24.9	21.8	17.6	14.1
	2%	DB	12.7	14.1	17.4	21.9	25.3	28.3	31.9	32.9	30.3	25.8	21.0	15.7
		MCWB	9.3	10.8	12.6	15.4	17.8	22.7	25.7	26.0	24.6	21.1	17.0	12.1
	5%	DB	11.6	12.8	15.9	20.2	24.0	27.1	31.0	32.0	29.2	24.7	19.9	14.7
		MCWB	8.2	9.6	11.6	14.5	17.5	22.0	25.5	25.8	24.0	20.0	16.1	11.1
	10%	DB	10.7	11.5	14.6	18.9	22.8	25.8	30.2	31.2	28.3	23.7	18.9	13.6
		MCWB	7.4	8.2	10.9	14.0	17.2	21.0	25.2	25.6	23.6	19.0	15.3	10.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	13.9	15.4	18.9	21.4	25.2	27.2	27.5	26.4	23.9	19.6	15.3
		MCDB	13.6	15.8	17.8	21.4	23.7	28.8	30.8	31.5	29.5	25.9	21.1	16.3
	2%	WB	10.4	11.9	13.6	17.2	20.1	24.2	26.6	26.9	25.6	22.5	18.1	13.1
		MCDB	11.9	13.6	15.9	19.9	22.4	27.1	30.1	30.8	28.7	25.0	20.0	14.8
	5%	WB	8.9	10.2	12.6	16.1	19.2	23.4	26.1	26.5	25.0	21.1	17.0	11.9
		MCDB	10.9	12.0	15.1	18.9	21.8	25.8	29.7	30.3	28.1	23.8	19.0	13.9
	10%	WB	7.7	8.8	11.4	15.1	18.5	22.5	25.6	26.1	24.5	19.8	15.8	10.7
		MCDB	10.2	11.1	14.1	17.8	21.4	24.7	29.2	29.8	27.5	22.7	18.4	13.2
Mean Daily Temperature Range	5% DB	MDBR	4.3	5.0	5.6	6.0	5.9	4.9	4.8	5.0	4.9	5.3	5.0	4.6
		MCDBR	5.6	6.7	7.4	8.1	7.7	6.6	5.8	5.8	5.4	6.0	5.7	5.6
		MCWBR	4.3	4.8	4.9	4.3	3.7	2.9	2.2	2.2	2.5	3.2	4.0	4.3
	5% WB	MCDBR	5.3	5.7	6.3	6.4	5.8	5.2	5.3	5.2	4.5	4.8	4.9	4.9
		MCWBR	5.0	5.3	5.2	4.2	3.7	2.8	2.2	2.2	2.4	3.4	4.1	4.7
Clear-Sky Solar Irradiance	taub	0.429	0.470	0.533	0.555	0.559	0.574	0.572	0.534	0.502	0.457	0.431	0.421	
	taud	2.060	1.987	1.839	1.840	1.890	1.940	1.992	2.098	2.137	2.160	2.145	2.095	
	Ebn at Noon	745	759	744	752	753	740	739	761	765	765	741	726	
	Edh at Noon	132	158	198	206	199	189	178	158	145	131	121	121	
All-Sky Solar Radiation	RadAvg	2.14	2.87	3.96	4.82	5.26	4.54	4.90	5.13	4.07	3.44	2.54	1.99	
	RadStd	0.19	0.28	0.29	0.41	0.60	0.49	0.74	0.57	0.41	0.36	0.23	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page